

Junyi (Conny) Zhou

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EDUCATION

Harvard University

Boston, MA

M.S. in Health Data Science

August 2025 – March 2027

- Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
- Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Agents Optimization

Emory University, College of Arts and Sciences

Atlanta, GA

B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925

August 2021 – May 2025

- Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures and Algorithms, Database Systems, Operating Systems, Computer Architecture

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL, Bash/Shell, Java, C++, R

Machine Learning & Scientific Computing: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn

Agentic AI & LLM Systems: OpenAI/Claude API, RAG, Prompt/Context Engineering, Agent Harnesses, Tool Calling, MCP, Multi-Agent Orchestration, Agent Evaluation, Human-in-the-Loop Systems

Full-Stack & Databases: React, Flask, REST APIs, Auth0, MongoDB, MongoDB Atlas Vector Search

Cloud & DevOps: AWS (AWS Machine Learning Associate Certified), Docker, Kubernetes, GitHub Actions

Developer Tools & Compute: Git, Cursor, Polar Browser, Grok Bot, Linux, Jupyter, SLURM/HPC, pytest, unittest

RESEARCH & ENGINEERING

Cross-Species Single-Cell Translation with Closed-Loop Autoresearch

Boston, MA

Graduate Researcher — Mooney Lab, Wyss Institute at Harvard

February 2026 – Present

GitHub: github.com/JunyiZhou-Conny/speciesOT

github.com/JunyiZhou-Conny/scgen-cellot-autoresearch

- Built an **end-to-end ML** mouse-to-human single-cell translation platform by adapting and benchmarking two leading *Nature Methods* models—CellOT (Bunne et al., 2023) and scGen (Lotfollahi et al., 2019)—using held-out cell types, decoded-space MMD gap closure, per-gene R^2 , and Jensen–Shannon divergence.
- Refactored notebook-driven research into a modular **Python package and CLI** with pluggable model/evaluation interfaces, **15** declarative YAML experiment specifications, backed AnnData processing for a **43-GB** atlas, and SHA-256 gene-axis checks that prevent silent feature misalignment.
- Designed a **closed-loop SLURM orchestrator** on Harvard FASRC Cannon—submit, monitor, triage, reflect, and adapt—that ran **338** experiments (318 completed; 20 failures handled) over **79.3 compute-hours** using single-factor ablations, checkpoint/resume, and fair-share-aware CPU routing.

Human-Governed Browser Agent System

Boston, MA

Software Developer, Sole Author

July 2026 – Present

GitHub: github.com/JunyiZhou-Conny/job-search-2026-2027-starter

- Built a human-governed browser-agent control plane for job discovery and application execution, using **GitHub** for durable policy and evidence, Google Sheets for runtime checkpoints, and JobrightAI as the application ledger across Polar local workflows and **Cursor cloud agents and Grok Bots**.
- Compiled YAML policy into bounded action sheets and one auditable runtime; enforced identity and work-authorization checks, evidence-grounded writing, duplicate guards, one-submit semantics, explicit unknown states, and restart-safe recovery.
- Evaluated the system across real Ashby, Greenhouse, Workday, SmartRecruiters, and SAP SuccessFactors workflows; added **200+** unit tests covering policy compilation, queue APIs, ledger integrity, and failure recovery.

RAG-Enabled Pediatric Airway Simulation & Debriefing Platform

Atlanta, GA

Full-Stack Engineer & Scrum Master — Emory University

January 2024 – August 2025

GitHub: github.com/JunyiZhou-Conny/Airway-Management-Assistant

- Led a six-person team, in collaboration with pediatric clinicians at Children’s Healthcare of Atlanta, to build a clinical training platform with a **React/JavaScript** frontend, **Flask REST API**, **Auth0 role-based access**, and **MongoDB** persistence for simulation cases, participant sessions, and debriefing histories.
- Grounded OpenAI-generated simulation dialogue and debriefing in clinician-authored cases and scanned medical references through **RAG** and **MongoDB Atlas Vector Search**, supporting Rapid Cycle Deliberate Practice through repeated pediatric bag-mask ventilation scenarios and immediate feedback.
- Deployed the **Dockerized Flask** backend on **AWS Elastic Beanstalk** and the **React** frontend through **S3/CloudFront** with **Route 53** and **HTTPS**; co-authored a **SAEM 2026** lightning oral abstract and an Emory/CHOA feasibility-study poster on the platform.